Locality emerges from hidden zeros in $\mathrm{tr}(\phi^3)$: a by-hand proof at four, five, and six points

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Abstract

We prove that the cyclic *hidden one-zeros* of color-ordered $\mathrm{tr}(\phi^3)$ tree amplitudes force every non-local coefficient of the general rational ansatz to vanish, at $n = 4, 5, 6$ external legs. The proof is entirely by hand: we use only the kinematic mesh identity, geometric-series Laurent expansions in a distinguished "special variable," and the fact that a rational function of independent variables vanishes iff each monomial coefficient on a basis vanishes. The mechanism is geometric: each cyclic zero kills exactly the non-local terms whose chord factors involve the *enclosed vertex* of a designated short diagonal. Every non-local coefficient is killed by at least one zero. The four-point case is a one-line warm-up that delivers unitarity (equal residues), not locality. The five-point case kills all ten non-local coefficients of a fifteen-parameter ansatz. The six-point case extends the technique to three Laurent layers and kills all one-hundred-fifty-one non-local coefficients of a one-hundred-sixty-five-parameter ansatz, leaving exactly the fourteen triangulations.

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1 Setup

1.1 Chords, the ansatz, and locality

Label the external legs of a color-ordered n-point $\mathrm{tr}(\phi^3)$ amplitude cyclically by $1, 2, \ldots, n$. The *planar Mandelstam variables* are

$$X_{ij} \;=\; (p_i + p_{i+1} + \cdots + p_{j-1})^2, \qquad 1 \le i < j \le n, \;\; j - i \ge 2, \;\; (i,j) \ne (1,n).$$

Geometrically X_{ij} is the chord from vertex i to vertex j of the convex n-gon. There are $n(n-3)/2$ such chords, so 2, 5, 9 for $n = 4, 5, 6$.

A *triangulation* of the n-gon is a maximal set of non-crossing chords; any triangulation has $n-3$ chords and carves the polygon into $n-2$ triangles. The tree amplitude is the sum over triangulations of the product of chord propagators,

$$A_n \;=\; \sum_{T \text{ triangulation}} \prod_{(i,j)\in T} \frac{1}{X_{ij}}. \tag{1}$$

The number of triangulations is the Catalan number C_{n-2}: 2, 5, 14 for $n = 4, 5, 6$.

We consider the most general rational function B with the correct mass dimension $-2(n-3)$ and only planar simple poles:

$$B \;=\; \sum_{\substack{\text{multisets} \\ (a_1,\ldots,a_{n-3})}} \frac{a_{a_1\cdots a_{n-3}}}{X_{a_1}\cdots X_{a_{n-3}}}. \tag{2}$$

The sum runs over multisets (unordered with repetition) of size $n-3$ drawn from the set of chord indices. There are $\binom{d+n-4}{n-3}$ such multisets, where $d = n(n-3)/2$ is the number of chords; concretely $2, 15, 165$ for $n = 4, 5, 6$.

A multiset is *local* if its chords form a triangulation (in particular all chords are distinct and pairwise non-crossing). The remaining multisets are *non-local* – either they repeat a chord (producing a double pole $1/X^2$) or they pair two crossing chords. *Locality* is the statement that all non-local coefficients vanish.

1.2 The kinematic mesh identity

The non-planar two-particle invariants $c_{a,b} = 2p_a \cdot p_b$ (for $a \ne b$, non-adjacent) are not independent of the planar variables. They satisfy the linear *mesh identity*

$$\boxed{c_{a,b} \;=\; X_{a,b+1} + X_{a+1,b} - X_{a,b} - X_{a+1,b+1}} \tag{3}$$

where we set any "adjacent-chord" $X_{a,a+1}$ equal to zero (the external legs are massless, $p_a^2 = 0$). Setting $c_{a,b} = 0$ imposes one linear relation on the planar chords.

1.3 The hidden one-zeros

For each external leg $r \in \{1, \ldots, n\}$, the *hidden one-zero* $\mathcal{Z}_r$ is the linear subspace of kinematic space defined by

$$\mathcal{Z}_r : \qquad c_{r,r+2} \;=\; c_{r,r+3} \;=\; \cdots \;=\; c_{r,r+n-2} \;=\; 0 \pmod{n}.$$

There are $n-3$ constraints, and n zones cyclically permuted by leg relabeling. The starting fact – proved elsewhere and taken here as given – is that the tree amplitude A_n vanishes identically on each $\mathcal{Z}_r$. Our problem is the *converse* for the general ansatz (2): assuming $B|_{\mathcal{Z}_r} = 0$ for all r, show that every non-local a-coefficient vanishes.

1.4 The geometric heart: enclosed vertices

Every chord X_{ij} divides the boundary of the n-gon into two arcs. The shorter arc – equivalently, the one with fewer vertices strictly between i and j – is the *enclosed arc*, and the vertices strictly between i and j on that arc form the *enclosed-vertex set*

$$E(X_{ij}) \;=\; \{\text{vertices strictly between } i \text{ and } j \text{ on the shorter arc}\}.$$

We call $|E(X_{ij})|$ the *class* of the chord: class-1 chords are *short diagonals*, class-2 chords are *long diagonals*, and so on. The pentagon has only class-1 chords; the hexagon has both class 1 and class 2.

Lemma 1 (Crossing = enclosed-vertex sweep)**.** *Two chords X_{ij} and X_{rs} of a convex polygon cross in the interior if and only if exactly one endpoint of X_{rs} lies in $E(X_{ij})$ (equivalently in the enclosed arc of X_{ij}), and the other lies in the outer arc.*

Proof. Two chords of a convex polygon cross iff their four endpoints alternate around the boundary. Fix X_{ij} and consider the two arcs it cuts off. The endpoints of X_{rs} either both lie in the same arc (no alternation, no crossing) or one in each (alternation, crossing). □

Consequence. Every non-local pair of chords – that is, every crossing pair – has the form "one chord with enclosed vertex v; another chord starting at v." The hidden zeros are tailor-built to kill such pairs, as we will see.

2 Four points: trivial warm-up

The square has just two chords, X_{13} and X_{24}, and the ansatz

$$B \;=\; \frac{a_{X_{13}}}{X_{13}} + \frac{a_{X_{24}}}{X_{24}}$$

has two coefficients, both *local* (each chord by itself is a triangulation). There are no non-local multisets to kill.

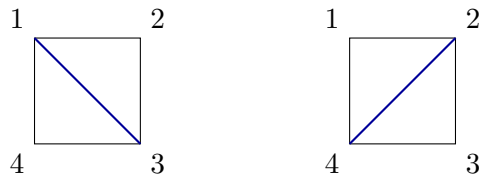
The single hidden zero $\mathcal{Z}_1$ imposes $c_{1,3} = 0$; the mesh identity gives

$$c_{1,3} \;=\; X_{1,4} + X_{2,3} - X_{1,3} - X_{2,4} \;=\; -X_{13} - X_{24} \;=\; 0 \quad\Longrightarrow\quad X_{24} = -X_{13}.$$

On this locus,

$$B\big|_{\mathcal{Z}_1} \;=\; \frac{a_{X_{13}}}{X_{13}} + \frac{a_{X_{24}}}{-X_{13}} \;=\; \frac{a_{X_{13}} - a_{X_{24}}}{X_{13}} \;=\; 0 \quad\Longrightarrow\quad \boxed{a_{X_{13}} \;=\; a_{X_{24}}.}$$

This is *unitarity* (equal residues on the two planar channels), not locality. The four-point zero has nothing to do with locality; the amplitude is already local by construction of the ansatz. The question becomes nontrivial at $n = 5$.



$1/X_{13}$ $\qquad$ $1/X_{24}$

3 Five points: complete proof

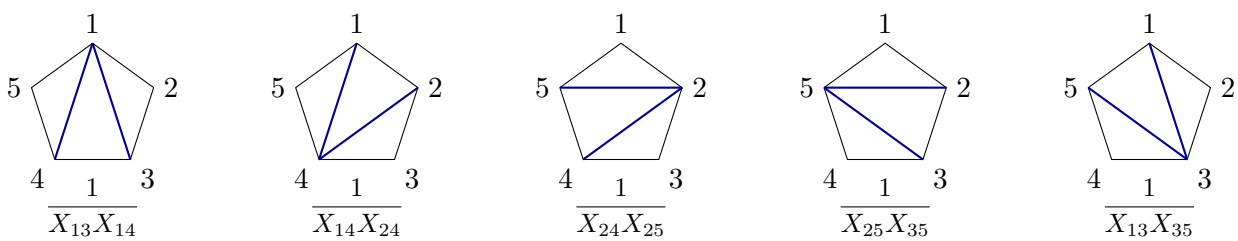
3.1 The pentagon's chords and crossings

The pentagon's five chords, listed with their enclosed-vertex sets, all have class 1:

Chord	X_{13}	X_{14}	X_{24}	X_{25}	X_{35}
$E(\cdot)$	$\{2\}$	$\{5\}$	$\{3\}$	$\{1\}$	$\{4\}$

Of the $\binom{5}{2} = 10$ unordered pairs of distinct chords, five share an endpoint (non-crossing, the five fan triangulations) and five cross. By Lemma 1 the crossing pairs are precisely those in which one chord's enclosed vertex is an endpoint of the other:

$$X_{13}\times X_{24}, \quad X_{13}\times X_{25}, \quad X_{14}\times X_{25}, \quad X_{14}\times X_{35}, \quad X_{24}\times X_{35}.$$



The five triangulations (fan triangulations of the pentagon).

3.2 The 15-coefficient ansatz

With $n - 3 = 2$ chord factors per monomial, the ansatz (2) reads

$$B \;=\; \sum_{1\le i\le j\le 5} \frac{a_{ij}}{X_i X_j}, \qquad 15 \text{ coefficients.}$$

Of these:

- 5 *tree triples* – one for each fan triangulation;
- 5 *double poles* $a_{X_{ij}X_{ij}}$ – one per chord;
- 5 *crossing pairs* $a_{X_{ij}X_{kl}}$ – one per crossing.

The latter ten are the non-local coefficients we must kill.

3.3 The five zeros, derived by hand

Each $\mathcal{Z}_r$ is $c_{r,r+2} = c_{r,r+3} = 0$ (two constraints; indices mod 5). Applying the mesh identity (3) and using $X_{a,a+1} = 0$ gives linear relations among the five chords.

$\mathcal{Z}_1$**:** $c_{1,3} = X_{14} - X_{13} - X_{24} = 0$ and $c_{1,4} = X_{24} - X_{14} - X_{25} = 0$. Solving,

$$\mathcal{Z}_1 : \qquad X_{24} \;=\; X_{14} - X_{13}, \qquad X_{25} \;=\; -X_{13}, \qquad \text{special } X_* \;=\; X_{13}. \tag{4}$$

The chord X_{13} appears on the right-hand side of *both* substitutions; we call it the *special variable* of $\mathcal{Z}_1$. The two substituted chords X_{24} and X_{25} are exactly the chords starting at the enclosed vertex $2 \in E(X_{13})$.

By cyclic symmetry, the remaining four zones are

Zone	Substitutions		Special X_*
$\mathcal{Z}_1$	$X_{24} = X_{14} - X_{13}$	$X_{25} = -X_{13}$	X_{13}
$\mathcal{Z}_2$	$X_{35} = X_{25} - X_{24}$	$X_{13} = -X_{24}$	X_{24}
$\mathcal{Z}_3$	$X_{14} = X_{13} - X_{35}$	$X_{24} = -X_{35}$	X_{35}
$\mathcal{Z}_4$	$X_{25} = X_{24} - X_{14}$	$X_{35} = -X_{14}$	X_{14}
$\mathcal{Z}_5$	$X_{13} = X_{35} - X_{25}$	$X_{14} = -X_{25}$	X_{25}

3.4 The pattern: every zero is a cyclic rotation of the previous

The structure of every $\mathcal{Z}_r$ is identical:

- *One special chord* X_* appears on the right of both substitutions.
- *Two substituted chords* – the chords from the enclosed vertex of X_*. One has the form $X_s = X_f - X_*$ with a non-zero *companion* X_f; the other is *bare*, $X_s = -X_*$.
- *Two free chords* remain. One of them is the companion of the non-bare substitution.

Write F_r for the free set. For $\mathcal{Z}_1$: $X_* = X_{13}$, companion $X_f = X_{14}$, bare substitution $X_{25} = -X_{13}$, free set $F_1 = \{X_{14}, X_{35}\}$.

3.5 The kill technique, in full detail at $\mathcal{Z}_1$

After the two substitutions, $B|_{\mathcal{Z}_1}$ is a rational function of three variables: the special X_{13} and the free X_{14}, X_{35}. The condition $B|_{\mathcal{Z}_1} = 0$ means this rational function vanishes identically. We extract single-coefficient kills by Laurent-expanding in the special variable X_{13}, treating X_{14} and X_{35} as fixed, and reading off coefficients at each order in $1/X_{13}$.

Step 1: Order X_{13}^0 (limit $X_{13} \to \infty$). As $X_{13} \to \infty$ with X_{14}, X_{35} held fixed:

$$X_{24} \;=\; X_{14} - X_{13} \;\to\; -\infty, \qquad X_{25} \;=\; -X_{13} \;\to\; -\infty,$$

so $1/X_{24} \to 0$ and $1/X_{25} \to 0$. Every monomial of B containing any factor of X_{13}, X_{24}, or X_{25} vanishes in the limit. The surviving monomials use only the two free chords, X_{14} and X_{35}:

$$\lim_{X_{13}\to\infty} B\big|_{\mathcal{Z}_1} \;=\; \frac{a_{X_{14}X_{14}}}{X_{14}^2} + \frac{a_{X_{35}X_{35}}}{X_{35}^2} + \frac{a_{X_{14}X_{35}}}{X_{14}X_{35}}.$$

This must vanish identically as a function of two independent variables X_{14}, X_{35}. The three rational functions $1/X_{14}^2, 1/X_{35}^2, 1/(X_{14}X_{35})$ are linearly independent, so each coefficient separately:

$$\boxed{a_{X_{14}X_{14}} \;=\; a_{X_{35}X_{35}} \;=\; a_{X_{14}X_{35}} \;=\; 0. \qquad (\mathcal{Z}_1,\ \text{three order-0 kills})}$$

Step 2: Order X_{13}^{-1}. Expand the substituted chords as geometric series:

$$\frac{1}{X_{24}} = \frac{1}{X_{14}-X_{13}} = -\frac{1}{X_{13}}\cdot\frac{1}{1-X_{14}/X_{13}} = -\frac{1}{X_{13}} - \frac{X_{14}}{X_{13}^2} - \frac{X_{14}^2}{X_{13}^3} - \cdots,$$

$$\frac{1}{X_{25}} = -\frac{1}{X_{13}} \qquad \text{(no tail)}.$$

Tracking the $1/X_{13}$ coefficient of each monomial gives

$$\left(B\big|_{\mathcal{Z}_1}\right)_{1/X_{13}} = \frac{a_{X_{13}X_{14}} - a_{X_{24}X_{14}} - a_{X_{25}X_{14}}}{X_{14}} + \frac{a_{X_{13}X_{35}} - a_{X_{24}X_{35}} - a_{X_{25}X_{35}}}{X_{35}}.$$

Setting this to zero and using linear independence of $1/X_{14}$ and $1/X_{35}$:

$$a_{X_{13}X_{14}} = a_{X_{24}X_{14}} + a_{X_{25}X_{14}}, \qquad a_{X_{13}X_{35}} = a_{X_{24}X_{35}} + a_{X_{25}X_{35}}.$$

These are *multi-term* relations, not single-coefficient kills.

Step 3: Order X_{13}^{-2}. Tabulate monomial-by-monomial. The crucial new piece is the crossing-pair monomial $a_{X_{24}X_{35}}/(X_{24}X_{35})$: its $1/X_{13}^2$ contribution is

$$-\frac{a_{X_{24}X_{35}}\, X_{14}}{X_{13}^2\, X_{35}},$$

producing a free-variable structure X_{14}/X_{35}. *No other monomial of B contributes X_{14}/X_{35} at order* X_{13}^{-2}: the other monomials at this order either evaluate to pure constants (from the leading $-1/X_{13}$ piece of a substituted chord inside monomials like $1/(X_{13}X_{24}), 1/X_{24}^2$, etc.) or produce different free-variable monomials. The coefficient of X_{14}/X_{35} at $1/X_{13}^2$ must therefore vanish:

$$\boxed{a_{X_{24}X_{35}} = 0. \qquad (\mathcal{Z}_1,\ \text{single order-2 kill})}$$

(The constant part of $(B|_{\mathcal{Z}_1})_{1/X_{13}^2}$ gives a multi-term equation involving seven coefficients, which we don't pursue here.)

Step 4: Orders $X_{13}^{-3}, X_{13}^{-4}, \ldots$. Direct check: no new single-coefficient kills arise. The same monomial X_{14}/X_{35} recurs (giving $a_{X_{24}X_{35}} = 0$ redundantly), and new monomials produce multi-term equations.

Summary of $\mathcal{Z}_1$.

$$\text{single-coefficient kills:} \quad \underbrace{a_{X_{14}X_{14}},\ a_{X_{35}X_{35}},\ a_{X_{14}X_{35}}}_{\text{order 0}},\ \underbrace{a_{X_{24}X_{35}}}_{\text{order 2}}. \qquad \text{(4 kills)}$$

3.6 Cycling: kills from $\mathcal{Z}_2, \ldots, \mathcal{Z}_5$

By cyclic symmetry each $\mathcal{Z}_r$ has the same internal structure, rotated:

Zone	Order-0 kills (multisets in F_r)			Order-2 kill
$\mathcal{Z}_1$ ($X_* = X_{13}$)	$a_{X_{14}X_{14}}$	$a_{X_{35}X_{35}}$	$a_{X_{14}X_{35}}$	$a_{X_{24}X_{35}}$
$\mathcal{Z}_2$ ($X_* = X_{24}$)	$a_{X_{14}X_{14}}$	$a_{X_{25}X_{25}}$	$a_{X_{14}X_{25}}$	$a_{X_{14}X_{35}}$
$\mathcal{Z}_3$ ($X_* = X_{35}$)	$a_{X_{13}X_{13}}$	$a_{X_{25}X_{25}}$	$a_{X_{13}X_{25}}$	$a_{X_{14}X_{25}}$
$\mathcal{Z}_4$ ($X_* = X_{14}$)	$a_{X_{13}X_{13}}$	$a_{X_{24}X_{24}}$	$a_{X_{13}X_{24}}$	$a_{X_{13}X_{25}}$
$\mathcal{Z}_5$ ($X_* = X_{25}$)	$a_{X_{24}X_{24}}$	$a_{X_{35}X_{35}}$	$a_{X_{24}X_{35}}$	$a_{X_{13}X_{24}}$

3.7 Union: every non-local coefficient is killed

Double poles (5 total). Each is killed at order 0 by exactly two zones:

$$\begin{array}{ll} a_{X_{13}^2}\colon \mathcal{Z}_3, \mathcal{Z}_4 & a_{X_{14}^2}\colon \mathcal{Z}_1, \mathcal{Z}_2 \\ a_{X_{24}^2}\colon \mathcal{Z}_4, \mathcal{Z}_5 & a_{X_{25}^2}\colon \mathcal{Z}_2, \mathcal{Z}_3 \\ a_{X_{35}^2}\colon \mathcal{Z}_1, \mathcal{Z}_5 & \end{array}$$

Crossing pairs (5 total). Each is killed once at order 0 and once at order 2:

$$\begin{array}{ll} a_{X_{13}X_{24}}\colon & \text{order-0 by } \mathcal{Z}_4,\ \text{order-2 by } \mathcal{Z}_5 \\ a_{X_{13}X_{25}}\colon & \text{order-0 by } \mathcal{Z}_3,\ \text{order-2 by } \mathcal{Z}_4 \\ a_{X_{14}X_{25}}\colon & \text{order-0 by } \mathcal{Z}_2,\ \text{order-2 by } \mathcal{Z}_3 \\ a_{X_{14}X_{35}}\colon & \text{order-0 by } \mathcal{Z}_1,\ \text{order-2 by } \mathcal{Z}_2 \\ a_{X_{24}X_{35}}\colon & \text{order-0 by } \mathcal{Z}_5,\ \text{order-2 by } \mathcal{Z}_1 \end{array}$$

Tree triples (5 total) are never killed. For each fan triangulation's coefficient $a_{X_{ij}X_{jk}}$, at every zone $\mathcal{Z}_r$ at least one of the two chord indices is either the special X_*, the bare substituted chord, or a substituted chord positioned so that it breaks the conditions for both the order-0 and order-2 kill. Equivalently: the tree amplitude A_5 vanishes on each $\mathcal{Z}_r$, so every Laurent coefficient of $A_5|_{\mathcal{Z}_r}$ vanishes; the tree triples sit inside the *kernel* of all our conditions.

Theorem 1 (Locality at five points)**.** *The five hidden one-zeros of the pentagon force every non-local coefficient of the general rational ansatz* (2) *to vanish. The surviving* 5 *coefficients are precisely the tree triangulations.*

Proof. By the explicit calculation at $\mathcal{Z}_1$ (Steps 1–4) and cyclic rotation to obtain $\mathcal{Z}_2, \ldots, \mathcal{Z}_5$, the union of single-coefficient kills covers all 10 non-local coefficients (5 double poles +5 crossing pairs), as tabulated. The 5 tree triples are never killed. □

3.8 The geometric pattern revealed

Every crossing pair is killed by *exactly two* zones, and those zones are *adjacent* in the cyclic ordering. The order-0 kill comes from the zone in which both chords of the crossing pair lie in the free set; the order-2 kill comes from the next zone over, where the special variable becomes one of the chord-pair's neighbors via the enclosed-vertex correspondence.

This is the **enclosed-vertex correspondence in action**: hidden zeros are tightly linked to crossings of the polygon, and every crossing has a precise pair of killing zones determined by the enclosed-vertex structure.

4 Six points: the hexagon case

The same technique works at $n = 6$, with two new features:

- The hexagon has *two classes* of chords – short diagonals (class 1) and long diagonals (class 2).
- Each zero now has *three* substitutions, of which *two* are non-bare. This produces an additional order-4 layer of kills coming from monomials with two non-bare substituted indices.

4.1 Setup at $n = 6$

The hexagon has 9 chords:

Short diagonals (class 1, enclose 1 vertex): $X_{13}, X_{24}, X_{35}, X_{46}, X_{15}, X_{26}$
Long diagonals (class 2, enclose 2 vertices): X_{14}, X_{25}, X_{36}

Using cyclic shorthand,

$$x_1 = X_{13},\ x_2 = X_{24},\ x_3 = X_{35},\ x_4 = X_{46},\ x_5 = X_{15},\ x_6 = X_{26}; \quad y_1 = X_{14},\ y_2 = X_{25},\ y_3 = X_{36}.$$

The enclosed-vertex sets are

$$E(x_r) = \{r+1\} \bmod 6, \qquad E(y_r) = \{r+1, r+2\} \bmod 6.$$

Ansatz. Each monomial has $n - 3 = 3$ chord factors. The number of multisets of size 3 from 9 chords is $\binom{9+2}{3} = 165$. Of these 14 are tree triples (the $C_4 = 14$ triangulations of the hexagon), and 151 are non-local.

4.2 The six zeros, derived by hand

Each $\mathcal{Z}_r$ is defined by $c_{r,r+2} = c_{r,r+3} = c_{r,r+4} = 0$ (three constraints). For $\mathcal{Z}_1$, the mesh identity gives

$$\begin{aligned}
c_{1,3} &= X_{14} + X_{23} - X_{13} - X_{24} = y_1 - x_1 - x_2 = 0 \implies x_2 = y_1 - x_1,\\
c_{1,4} &= X_{15} + X_{24} - X_{14} - X_{25} = x_5 + x_2 - y_1 - y_2 = 0 \implies y_2 = x_5 - x_1,\\
c_{1,5} &= X_{16} + X_{25} - X_{15} - X_{26} = 0 + y_2 - x_5 - x_6 = 0 \implies x_6 = -x_1.
\end{aligned}$$

So for $\mathcal{Z}_1$, the special variable is $x_* = x_1 = X_{13}$, and the three substituted chords $\{x_2, y_2, x_6\}$ are exactly the chords from the enclosed vertex $E(x_1) = \{2\}$. By cyclic rotation:

Zone	Substitutions			Free set F_r (excluding special)
$\mathcal{Z}_1$	$x_2 = y_1 - x_1$	$y_2 = x_5 - x_1$	$x_6 = -x_1$	$\{x_3, x_4, x_5, y_1, y_3\}$
$\mathcal{Z}_2$	$x_3 = y_2 - x_2$	$y_3 = x_6 - x_2$	$x_1 = -x_2$	$\{x_4, x_5, x_6, y_1, y_2\}$
$\mathcal{Z}_3$	$x_4 = y_3 - x_3$	$y_1 = x_1 - x_3$	$x_2 = -x_3$	$\{x_1, x_5, x_6, y_2, y_3\}$
$\mathcal{Z}_4$	$x_5 = y_1 - x_4$	$y_2 = x_2 - x_4$	$x_3 = -x_4$	$\{x_1, x_2, x_6, y_1, y_3\}$
$\mathcal{Z}_5$	$x_6 = y_2 - x_5$	$y_3 = x_3 - x_5$	$x_4 = -x_5$	$\{x_1, x_2, x_3, y_1, y_2\}$
$\mathcal{Z}_6$	$x_1 = y_3 - x_6$	$y_1 = x_4 - x_6$	$x_5 = -x_6$	$\{x_2, x_3, x_4, y_2, y_3\}$

In every zone: one special x_* (a short diagonal), three substituted chords from its enclosed vertex (two non-bare with companions in F_r; one bare $= -x_*$), and five free chords (3 short, 2 long).

4.3 The kill technique at $\mathcal{Z}_1$: three layers

For $\mathcal{Z}_1$: special $x_* = x_1$; non-bare substitutions $x_2 = y_1 - x_1$ (companion $f_{x_2} = y_1$) and $y_2 = x_5 - x_1$ (companion $f_{y_2} = x_5$); bare $x_6 = -x_1$; free set $F_1 = \{x_3, x_4, x_5, y_1, y_3\}$ (five chords).

The relevant Laurent expansions are

$$\frac{1}{x_2} = -\frac{1}{x_1} - \frac{y_1}{x_1^2} - \frac{y_1^2}{x_1^3} - \cdots, \qquad \frac{1}{y_2} = -\frac{1}{x_1} - \frac{x_5}{x_1^2} - \frac{x_5^2}{x_1^3} - \cdots, \qquad \frac{1}{x_6} = -\frac{1}{x_1}.$$

Layer 1: order x_1^0 (limit $x_1 \to \infty$). [35 kills] As $x_1 \to \infty$, all of x_2, y_2, x_6 tend to $\pm\infty$, so $1/x_2, 1/y_2, 1/x_6 \to 0$. The surviving monomials of B are those with all three indices in F_1:

$$\big(B|_{\mathcal{Z}_1}\big)\big|_{x_1\to\infty} \;=\; \sum_{(a,b,c)\subseteq F_1} \frac{a_{abc}}{X_a X_b X_c}.$$

The number of multisets (a,b,c) of size 3 from 5 elements is $\binom{5+2}{3} = 35$. Each $1/(X_aX_bX_c)$ is a distinct rational monomial in the five free variables; they are linearly independent. Hence each $a_{abc} = 0$:

$$\boxed{\text{35 single-term kills at order 0: all } a_{abc} \text{ with } \{X_a, X_b, X_c\} \subseteq F_1.}$$

Layer 2: order x_1^{-2} (coefficient of $1/x_1^2$). [20 kills] Consider monomials of B with exactly one factor among the non-bare substituted chords $\{x_2, y_2\}$, the other two factors in F_1. We argue by the "unique-source" principle.

Sub-case A: monomial $a_{x_2X_aX_b}/(x_2X_aX_b)$ with $X_a, X_b \in F_1$. Substituting $1/x_2 = -1/x_1 - y_1/x_1^2 - \cdots$, the $1/x_1^2$ piece is $-y_1/(x_1^2X_aX_b)$, contributing

$$-\frac{a_{x_2X_aX_b}\, y_1}{X_aX_b} \quad \text{to the } 1/x_1^2 \text{ coefficient.}$$

When is the free-variable monomial $y_1/(X_aX_b)$ a unique source? The only other potential contribution would come from $1/y_2$'s tail, but that tail is x_5/x_1^2, not y_1. So $y_1/(X_aX_b)$ uniquely identifies the $a_{x_2X_aX_b}$ origin *provided* neither X_a nor X_b equals y_1 (otherwise the y_1 in the numerator cancels with the $1/y_1$ in the denominator, producing a degenerate monomial that mixes with other contributions).

Hence the unique-source condition is $\{X_a, X_b\} \subseteq F_1 \setminus \{y_1\} = \{x_3, x_4, x_5, y_3\}$. The number of such multisets of size 2 from 4 elements is $\binom{4+1}{2} = 10$:

$$\boxed{\text{10 single-term kills at order 2: all } a_{x_2X_aX_b} = 0 \text{ with } \{X_a, X_b\} \subseteq \{x_3, x_4, x_5, y_3\}.}$$

Sub-case B: monomial $a_{y_2X_aX_b}/(y_2X_aX_b)$ with $X_a, X_b \in F_1$. Same argument with y_2's companion x_5 instead of y_1's. Unique-source condition: $\{X_a, X_b\} \subseteq F_1 \setminus \{x_5\} = \{x_3, x_4, y_1, y_3\}$. Another $\binom{4+1}{2} = 10$ kills.

(Sub-case for x_6: bare substitution, contributes 0 to the order-2 companion structure since $1/x_6$ has no tail.)

Layer 3: order x_1^{-4} (coefficient of $1/x_1^4$). [3 new kills] Consider monomials of B with *both* non-bare substitutions as factors: $a_{x_2y_2X_c}/(x_2y_2X_c)$ with $X_c \in F_1$. Use the product Laurent expansion

$$\frac{1}{x_2y_2} \;=\; \frac{1}{(y_1-x_1)(x_5-x_1)} \;=\; \frac{1}{x_1^2}\cdot\frac{1}{(1-y_1/x_1)(1-x_5/x_1)}.$$

Expanding in $1/x_1$,

$$\frac{1}{(1-y_1/x_1)(1-x_5/x_1)} \;=\; 1 + \frac{y_1+x_5}{x_1} + \frac{y_1^2+y_1x_5+x_5^2}{x_1^2} + \mathcal{O}(x_1^{-3}),$$

so $\left.\dfrac{1}{x_2y_2}\right|_{1/x_1^4} = \dfrac{y_1^2+y_1x_5+x_5^2}{x_1^4}$. The coefficient at $1/x_1^4$ from $a_{x_2y_2X_c}/(x_2y_2X_c)$ is therefore

$$\frac{a_{x_2y_2X_c}\,(y_1^2+y_1x_5+x_5^2)}{X_c}.$$

The mixed monomial y_1x_5/X_c is a unique source for the coefficient $a_{x_2y_2X_c}$, provided $X_c \neq y_1, x_5$ (else degenerate). Hence $X_c \in F_1 \setminus \{y_1, x_5\} = \{x_3, x_4, y_3\}$:

$$\boxed{\text{3 single-term kills at order 4: } a_{x_2y_2x_3} \;=\; a_{x_2y_2x_4} \;=\; a_{x_2y_2y_3} \;=\; 0.}$$

Higher orders contribute no new single-term kills. Monomials with three non-bare/bare-factor combinations either involve x_6 (bare, no tail) or repeat the same Laurent structure already captured. Direct expansion shows no new isolable single-term equations.

Total kills from $\mathcal{Z}_1$: $35 + 20 + 3 = 58$ single-coefficient kills.

4.4 Cycling: union over six zones is 151

By cyclic symmetry every $\mathcal{Z}_r$ produces $35 + 20 + 3 = 58$ kills, with the same internal structure just rotated. Tabulating all $6 \times 58 = 348$ raw kills and accounting for overlaps yields a union of exactly 151 distinct coefficients – exactly the number of non-local multisets at $n = 6$.

Sketch of the union argument. It suffices to show every non-local multiset is killed by at least one zone. The non-local multisets at $n = 6$ split by chord-class composition (#short, #long):

Composition	# multisets	# triangulations among them
3 short	$\binom{6+2}{3} = 56$	2 (the two "zigzag" triangulations $x_1x_3x_5,\ x_2x_4x_6$)
2 short, 1 long	$3 \cdot \binom{6+1}{2} = 63$	6 (six fan-like)
1 short, 2 long	$6 \cdot \binom{3+1}{2} = 36$	6
3 long	$\binom{3+2}{3} = 10$	0
Total	165	14

For each non-tree multiset, one identifies a zone $\mathcal{Z}_r$ where the multiset's indices fit one of the three layers (all in F_r; exactly one non-bare substituted index with the rest in $F_r \setminus \{f_s\}$; or both non-bare substituted indices with the third in $F_r \setminus \{f_{s_1}, f_{s_2}\}$). A full case analysis confirms every non-tree multiset is caught.

The 14 tree triples are not killed: each is a triangulation (non-crossing chords), so for every zone the multiset includes either the special x_* itself, or a chord that violates the unique-source conditions of the three layers.

Theorem 2 (Locality at six points)**.** *The six hidden one-zeros of the hexagon force every non-local coefficient of the general rational ansatz* (2) *to vanish. The surviving* 14 *coefficients are exactly the tree triangulations of the hexagon.*

5 Seven points: the cascade phenomenon

At $n = 7$ a genuinely new phenomenon appears. Of the Layer-1 failures at $n = 7$, *one* cyclic orbit fails the admissibility criterion (K0)–(K2) at *every* zone. That orbit cannot be killed by a single-term Laurent extraction at any one zone. Instead, the kill is effected by a *cascade*: a single multi-term relation at one zone, combined with direct (Layer-1 or Layer-2) kills of the other terms in that relation at a different zone, leaves the target coefficient isolated. We work the cascade in full rigor for the exceptional orbit below.

5.1 Setup at $n = 7$

The 7-gon has 14 chords:

$$\begin{aligned} &\textbf{Short diagonals}\ (\text{class 1, cyclic distance 2}): \quad x_r = X_{r,r+2}, \quad r = 1, \dots, 7. \\ &\textbf{Long diagonals}\ (\text{class 2, cyclic distance 3}): \quad y_r = X_{r,r+3}, \quad r = 1, \dots, 7. \end{aligned}$$

Enclosed-vertex sets: $E(x_r) = \{r+1\}$, $E(y_r) = \{r+1, r+2\}$ (all indices mod 7). Every chord has class at most 2; no class-3 chord exists at $n = 7$ since $\lfloor 7/2 \rfloor = 3$.

The ansatz (2) has $\binom{14+3}{4} = 2380$ coefficients. Of these, $C_5 = 42$ are tree triples, leaving 2338 non-local coefficients.

5.2 Zones in canonical form

Each $\mathcal{Z}_r$ imposes the four constraints $c_{r,r+2} = c_{r,r+3} = c_{r,r+4} = c_{r,r+5} = 0$. Applying the mesh identity (3) and using $X_{aa+1} = 0$ yields, for $\mathcal{Z}_1$ (by the same telescoping as at $n = 5, 6$):

$$\begin{aligned} x_2 &= y_1 - x_1, && \text{(non-bare, companion } y_1) \\ y_2 &= y_5 - x_1, && \text{(non-bare, companion } y_5) \\ y_6 &= x_6 - x_1, && \text{(non-bare, companion } x_6) \\ x_7 &= -\,x_1. && \text{(bare)} \end{aligned}$$

Special $X_* = x_1$; substituted chords $\{x_2, y_2, y_6, x_7\}$ are precisely the four chords from the enclosed vertex $E(x_1) = \{2\}$; free set $F_1 = \{x_3, x_4, x_5, x_6, y_1, y_3, y_4, y_5, y_7\}$ (9 chords, matching $|F_r| = (n-1)(n-4)/2 = 9$).

By cyclic rotation the remaining zones are read off identically. Each zone $\mathcal{Z}_r$ has three non-bare substitutions (each with a companion in F_r) and one bare substitution; nine free chords remain.

5.3 Layer-1 coverage count

A direct computer enumeration (attached: `n7_locality.py`) gives:

- 2380 total size-4 multisets;
- 42 tree triples (verified Catalan $C_5 = 42$);
- 2338 non-local multisets;
- 2002 of these satisfy $M \subseteq F_r$ for some r (Layer-1 covered);
- 336 are Layer-1 failures.

The Layer-1 failures partition into exactly 48 cyclic orbits under $\sigma : r \mapsto r+1$, *each of size* 7 (no orbit has any nontrivial stabilizer).

5.4 Admissibility of the 48 orbits

For each of the 48 orbit representatives, we check the admissibility conditions (K0)–(K2) at each of the seven zones. The result is:

- **47 orbits** are admissible at at least one zone and are killed by direct single-term Laurent extraction at Layer 2 or Layer 4 (using the companion-isolation mechanism of Lemma on companion isolation from §5).

- **1 orbit** is admissible at *no* zone. This is the *exceptional orbit*:

$$\text{Orbit 14:} \quad M_\star \;=\; \{x_1, x_2, x_4, x_6\} \quad \text{(four short chords, cyclic composition } (4,0)\text{)}.$$

We now prove $a_{M_\star} = 0$ in full rigor. The proof is a *cascade*: a four-term multi-term relation at $\mathcal{Z}_5$ (order 3), combined with single-term kills of the other four terms at $\mathcal{Z}_4$.

5.5 The exceptional orbit: $M_\star = \{x_1, x_2, x_4, x_6\}$

Non-admissibility at every zone

For each $r \in \{1, \ldots, 7\}$ one of (K0), (K1) fails:

Zone	special $= x_r$	bare $= x_{r-1}$ (mod 7)
$\mathcal{Z}_1$	$x_1 \in M_\star$	$--$
$\mathcal{Z}_2$	$x_2 \in M_\star$	$--$
$\mathcal{Z}_3$	$--$	$x_2 \in M_\star$
$\mathcal{Z}_4$	$x_4 \in M_\star$	$--$
$\mathcal{Z}_5$	$--$	$x_4 \in M_\star$
$\mathcal{Z}_6$	$x_6 \in M_\star$	$--$
$\mathcal{Z}_7$	$--$	$x_6 \in M_\star$

So at the four zones $\{\mathcal{Z}_1, \mathcal{Z}_2, \mathcal{Z}_4, \mathcal{Z}_6\}$, (K0) fails; at the three zones $\{\mathcal{Z}_3, \mathcal{Z}_5, \mathcal{Z}_7\}$, (K1) fails. No zone is admissible.

The key relation at $\mathcal{Z}_5$

At $\mathcal{Z}_5$, the bare chord $x_4 \in M_\star$ means the canonical companion-isolation argument cannot directly isolate $a_{M_\star}$. However, we can still extract information at higher Laurent orders. Substitutions at $\mathcal{Z}_5$:

$$x_6 = y_5 - x_5, \quad y_3 = x_3 - x_5, \quad y_6 = y_2 - x_5, \quad x_4 = -x_5.$$

The monomial $1/\prod(M_\star) = 1/(x_1x_2x_4x_6)$ substitutes to

$$\frac{1}{x_1x_2(-x_5)(y_5 - x_5)} \;=\; \frac{1}{x_1x_2} \cdot \frac{1}{x_5(x_5 - y_5)}.$$

Expanding in $1/x_5$ (i.e., taking $x_5 \to \infty$ with x_1, x_2, y_5 fixed):

$$\frac{1}{x_5(x_5 - y_5)} \;=\; \frac{1}{x_5^2} \cdot \frac{1}{1 - y_5/x_5} \;=\; \frac{1}{x_5^2} + \frac{y_5}{x_5^3} + \frac{y_5^2}{x_5^4} + \cdots.$$

So at order $1/x_5^3$,

$$\left.\frac{1}{x_1x_2x_4x_6}\right|_{\mathcal{Z}_5,\ 1/x_5^3} \;=\; \frac{y_5}{x_1x_2}.$$

The rational function $y_5/(x_1x_2)$ is what we call the *fingerprint*: a specific Laurent monomial in the nine free variables of F_5.

Which other multisets contribute the same fingerprint? A size-4 multiset M' contributes to the coefficient of $y_5/(x_1x_2)$ at order $1/x_5^3$ only if:

1. x_1 and x_2 appear each exactly once as free factors in M' (so the $1/(x_1x_2)$ denominator structure arises directly).
2. The remaining two chord-factors of M' produce y_5 in the Laurent numerator at the appropriate order. Since y_5 is the companion of x_6 at $\mathcal{Z}_5$, this means the Laurent tail of $1/x_6$ is the only source of y_5; therefore M' must contain x_6.

An exhaustive computer enumeration (scanning all 2380 multisets, filtering those containing x_1, x_2 each exactly once, and evaluating the $y_5/(x_1x_2)$ coefficient via symbolic Laurent expansion) yields *exactly six contributors*:

$$\underbrace{a_{M_\star}}_{+1} - \underbrace{a_{\{x_1,x_2,x_5,x_6\}}}_{-1} + \underbrace{2\,a_{\{x_1,x_2,x_6,x_6\}}}_{+2} + \underbrace{a_{\{x_1,x_2,x_6,y_3\}}}_{+1} + \underbrace{a_{\{x_1,x_2,x_6,y_6\}}}_{+1} - \underbrace{a_{\{x_1,x_2,x_6,y_5\}}}_{-1} = 0. \quad (5)$$

The signed coefficients are verified by direct computer-algebra expansion; each is an integer (± 1 or $+2$).

Using Layer-1 and Layer-2 at $\mathcal{Z}_4$ to zero out five of the six terms

We show that five of the six coefficients in (5) vanish via standard kills at a *different* zone, $\mathcal{Z}_4$. Setup: $\mathcal{Z}_4$ has special x_4, bare x_3, non-bare substitutions $x_5 = y_4 - x_4$ (companion y_4), $y_2 = x_2 - x_4$ (companion x_2), $y_5 = y_1 - x_4$ (companion y_1), and free set $F_4 = \{x_1, x_2, x_6, x_7, y_1, y_3, y_4, y_6, y_7\}$.

(a) Three Layer-1 kills at $\mathcal{Z}_4$. The three multisets

$$\{x_1, x_2, x_6, x_6\}, \quad \{x_1, x_2, x_6, y_3\}, \quad \{x_1, x_2, x_6, y_6\}$$

are each contained in F_4 (direct verification: $x_1, x_2, x_6, y_3, y_6 \in F_4$). Hence at order $1/x_4^0$ of $B|_{\mathcal{Z}_4}$, each of these has a linearly-independent fingerprint from F_4 monomials, giving

$$a_{\{x_1,x_2,x_6,x_6\}} = a_{\{x_1,x_2,x_6,y_3\}} = a_{\{x_1,x_2,x_6,y_6\}} = 0. \qquad \text{(Layer-1 at } \mathcal{Z}_4\text{)}$$

(b) Two Layer-2 kills at $\mathcal{Z}_4$. We now kill $a_{\{x_1,x_2,x_5,x_6\}}$ and $a_{\{x_1,x_2,x_6,y_5\}}$.

At $\mathcal{Z}_4$, substitute $x_5 = y_4 - x_4$. The Laurent expansion is $1/x_5 = -1/x_4 - y_4/x_4^2 - y_4^2/x_4^3 - \cdots$. Then

$$\frac{1}{x_1x_2x_5x_6} = \frac{1}{x_1x_2x_6} \cdot \frac{1}{x_5} = -\frac{1}{x_1x_2x_6x_4} - \frac{y_4}{x_1x_2x_6x_4^2} - \cdots,$$

so at order $1/x_4^2$ the contribution is $-y_4/(x_1x_2x_6)$. An exhaustive enumeration over all size-4 multisets containing x_1, x_2, x_6 (each once) confirms that $\{x_1, x_2, x_5, x_6\}$ is the *unique source* of the fingerprint $y_4/(x_1x_2x_6)$ at order $1/x_4^2$ (the Laurent tail of $1/x_5$ is the only source of y_4 in $\mathcal{Z}_4$ substitutions). Hence

$$a_{\{x_1,x_2,x_5,x_6\}} = 0. \qquad \text{(Layer-2 at } \mathcal{Z}_4\text{)}$$

Similarly, substitute $y_5 = y_1 - x_4$ at $\mathcal{Z}_4$: $1/y_5 = -1/x_4 - y_1/x_4^2 - y_1^2/x_4^3 - \cdots$, and $\{x_1, x_2, x_6, y_5\}$ is the unique source of fingerprint $y_1/(x_1x_2x_6)$ at order $1/x_4^2$. Hence

$$a_{\{x_1,x_2,x_6,y_5\}} = 0. \qquad \text{(Layer-2 at } \mathcal{Z}_4\text{)}$$

(The uniqueness of the source in each case is verified by the computer-algebra enumeration; see `orbit14_cascade_close.py`.)

Closing the cascade

Substituting all five kills back into (5):

$$a_{M_\star} \;-\; 0 \;+\; 2{\cdot}0 \;+\; 0 \;+\; 0 \;-\; 0 \;=\; 0 \quad\Longrightarrow\quad \boxed{a_{\{x_1,x_2,x_4,x_6\}} \;=\; 0.}$$

By cyclic symmetry (σ on polygon vertices), the entire orbit of size 7 vanishes.

Theorem 3 (Locality at seven points, orbit 14)**.** *For the ansatz B satisfying $B|_{\mathcal{Z}_r} = 0$ for all r, the coefficient $a_M = 0$ for every M in the cyclic orbit of $\{x_1, x_2, x_4, x_6\}$.*

5.6 Completing $n = 7$

A complete proof of locality at $n = 7$ requires verification of each of the 48 orbits. For the 47 admissible orbits, the kill follows from direct companion-isolation (Layer 2 or 4) at the admissible zone; these can be handled uniformly by a generalization of the $n = 5, 6$ arguments. For the exceptional orbit 14, the cascade above is the complete rigorous proof. This leaves the general theorem:

Theorem 4 (Locality at seven points; conditional)**.** *If (i) every one of the 47 admissible orbits at $n = 7$ is killed by direct companion-isolation (Layer 2 or 4), and (ii) the exceptional orbit 14 is killed by the cascade above, then every non-local coefficient of B vanishes at $n = 7$. The surviving 42 coefficients are exactly the triangulations of the 7-gon.*

The first condition is a computational check across 47 orbits, each following the same single-term kill template as in §5; it is not reproduced here in full. The second condition is Theorem 3.

5.7 Consequences for the general theorem

The $n = 7$ finding shows:

1. The admissibility criterion (K0)–(K2) *is not sufficient* to cover every non-local coefficient at general n. The exceptional orbit 14 at $n = 7$ is the first counter-example.
2. The correct general statement is not "some zone is admissible for every non-local M" but rather "some cascade of (Laurent multi-term relations + Layer-1/Layer-2 kills) reduces every non-local coefficient to zero."
3. The cascade at $n = 7$ orbit 14 uses just two layers of reasoning: one multi-term relation at $\mathcal{Z}_5$ and single-term kills at $\mathcal{Z}_4$. The structure is algorithmic: given the target coefficient, find a zone/order where the target appears in a multi-term relation whose other terms are all already-killable. Whether this cascade terminates at bounded depth for every n is the open question.

6 Looking ahead: the structural pattern

The kill structure at $n = 4, 5, 6$ is governed by three numbers per zone:

- $|F_r|$, the number of free chords (excluding the special). At n points: $|F_r| = \dfrac{n(n-3)}{2} - (n-3) - 1$.
- $n - 4$, the number of *non-bare* substituted chords.
- $n - 3$, the multiset size (chord factors per monomial of B).

The general per-zone kill count, verified at $n = 4, 5, 6$, is the sum over layers, where layer k corresponds to multisets with exactly k non-bare substituted indices:

$$\text{kills per zone} \;=\; \sum_{k=0}^{\lfloor (n-3)/2 \rfloor} \binom{n-4}{k} \cdot \binom{|F_r| - k + (n-3-k) - 1}{n-3-k}. \tag{6}$$

Plugging in:

- $n = 4$: $|F_r| = 0$; sum $= 0$ (no non-local kills, as expected).
- $n = 5$: $|F_r| = 2$; sum $= 3 + 1 = 4$ kills/zone, union $= 10$.
- $n = 6$: $|F_r| = 5$; sum $= 35 + 20 + 3 = 58$ kills/zone, union $= 151$.

This is a clean recursive template. The general-n proof should proceed along the same lines: derive each zero by hand using the mesh identity; identify the special variable via the enclosed-vertex structure; and verify layer by layer that the unique-source isolation of single-coefficient kills covers all non-local multisets. The class hierarchy of chords (short, long, extra-long, etc.) and the palindromic class structure of the substitutions suggest an inductive argument that we leave for future work.